



Trade Implications of Climate Change in United States Ports

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January 31, 2019
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WRIT 340

When Hurricane Katrina struck the Gulf Coast of the United States in 2005, the storm caused over \$100 million in damages to the commercial public port system of the State of Mississippi (PEER, 2006). In 2017, Hurricane Harvey caused between \$6-10 billion in damages to critical infrastructure in the Eleventh Federal Reserve District, which includes the entire port system of the State of Texas (Kaplan, 2017). Similar disasters are expected with increasing frequency and power due to the changing global climate. As sea levels rise and storms grow stronger, United States seaports could face operational delays amounting to billions of dollars per day (Becker et al., 2013). In time, the United States risks a significant decline in revenue generated from seafaring trade and overall GDP as seaports contend with the effects of climate change. While the politics of climate change remain a controversial subject in the United States, analysis of the phenomenon's effects on the supply and demand of the seafaring trade industry requires a coordinated policy response with precise and deliberate execution. The American public and its representatives in government must take action to prevent a contraction of the freight industry and reduction in national welfare and productivity.

Studies show American voters are polarized with respect to climate policy (Ehret, Van Boven, & Sherman, 2018). Regardless of their personal knowledge regarding climate change or its implications for the economy at large, voters rely on partisan cues to formulate policy evaluations and determine their votes. However, this was not always the case. A Gallup poll in 1989 found that Democrats and Republicans were equally worried about "global warming," with 67% of each group responding in the affirmative (Carmichael, Brulle, & Huxster, 2017). The same poll was repeated in each of the following decades, the partisan divide growing with each iteration. At the final polling in 2016, 84% of Democrats and just 40% of Republicans expressed

concern about climate change. Such polarization has led to a legacy of inaction in the face of climate change.

The United States' inability to overcome partisan politics in the climate arena has negative implications for the economic welfare of the nation's ports and trade infrastructure, as well as the general welfare of the nation. The United States' withdrawal from the Paris Agreement on emissions reductions, as well as the U.S. president's declaration of climate change as a "hoax" and his decision to halt payments to the United Nations' Green Climate Fund, weakened the nation's position as a leader on the international stage while simultaneously propelling rivals like China and Europe. Economists in 2017 predicted retaliatory carbon tariffs by rivals against imports from the United States, to help offset the cost to the rest of the world of covering for the United States' emissions (Bohringer and Rutherford, 2017). Whether or not such carbon tariffs are enacted, the United States' spurning of emissions reduction efforts contributes to increasing risk profiles surrounding ports and trade infrastructure.

The American people directly assume the increasing risk of seaport infrastructure operations. The United States seaport system accounts for \$4.6 trillion in economic activity per year and provides over 23 million jobs for American workers (Becker et al., 2013). The economic activity and jobs creation attributed to port-based trade are amplified even still when commerce in importing firms at the end of the supply chain is considered. Most consumer facing firms in the United States fall into this category. United States-based production has shifted largely to high-end manufacturing in recent years, including heavy equipment, aircraft, and microchip semiconductors (Manning, 2009). These goods typically never meet the consumer at market. Trade processed at port facilities include intermediate and final goods and commodities that generate commerce and jobs creation for consumer facing firms throughout the United

States. As climate change raises incidence rates of high-impact storms and generally increases the risks associated with seafaring trade, the costs of imported inputs for American goods will rise. With greater costs in their supply chains, American firms face prospects of decreased productivity through layoffs and capital reductions. American productivity, and subsequently GDP, face a significant decline when ports bear the economic damages associated with climate change.

As productivity declines, so too will general welfare. The Keynesian multiplier theory outlines the cycles of increased employment and prosperity that follow increases in government spending (Batini et al., 2014). When a government introduces money into a system, beneficiaries of that money will save and invest. They will consume goods and services. They will also be taxed. The Keynesian multiplier theory asserts that a dollar of spending by the federal government generates more than a dollar of economic growth. This theory can be generalized beyond the government to capital inflows from foreign investors and consumers. Capital inflow introduces money that can be saved, invested, and taxed. These new funds are then multiplied through financial institutions and government service provisions. The overall value of the American general welfare increases by more than the initial capital inflow. Nearly all of this growth is generated through port-based exports and imports. More than 80% of global merchandise trade by volume and 70% of trade by value is seaborne. As climate change increases the risk of seaborne trade, the cost of imports will increase and productivity will decrease. Exports and capital inflows will also fall, limiting the multiplier effect and reducing overall general welfare.

A study on the economic impacts of a tsunami in California modeled this welfare reduction process. The study found evidence of a rippling effect of GDP reduction (the multiplier

effect) due to cargo disruptions such as delays in shipments at the Port of Los Angeles and the Port of Long Beach. The twin ports, comprising the San Pedro Bay port complex, rank together as the ninth busiest port complex in the world (“Facts and Figures”). The Port of Los Angeles alone is the busiest container port in North America. The California tsunami study employed a variety of value reduction measures at the port level. These included estimates of the economic effects of “import and export value, facility downtime, cargo damages, repair and replace costs of damaged buildings, and gross output losses for marina-related sectors and the fishing industry” (Rose et al., 2016). The study measured value reduction over a two-day port shutdown, showing that a loss of operations for just two days would reduce the value of exports by \$132 million in Southern California alone. Beyond the port, value reduction would occur in downstream industries regionally and nationally through the multiplier effect. The study found that the economic spillover to the rest of the United States from a two-day port disruption in California would yield a \$10 billion reduction in national GDP. Controlling for national resilience efforts, a California tsunami would still reduce the United States’ GDP by more than half a billion dollars.

Beyond commerce-related seaport damages, port systems also face higher risk due to increasing populations in port cities. As port cities are often found in the most desirable coastal regions, these cities have seen some of the greatest rates of population growth nationally since 2010 (Klotzbach et al., 2018). Particularly hurricane-prone regions experienced more growth than anywhere else in America. According to a 2018 study, counties in Florida and Texas were among the fastest growing counties between 2010 and 2016. The West and the Coastal West (California, Oregon, and Washington) grew at annual rates of 3.9% and 2.1% respectively since 1970. At current rates, total population in port cities could more than triple by 2070 (Nicholls et

al., 2008). A port system's ability to contain damage is threatened by higher concentrations of people living in the surrounding area. With more people to manage in times of natural disaster, port and emergency management administrations are less able to focus on mitigating economic damages and streamlining import/export processes. Exposure rates to climate-related tropical storms will continue to rise as people move to port cities in hurricane-prone coastal regions.

Further, increased populations in port cities yield higher concentrations of wealth. A greater labor force earns higher incomes due to the laws of supply and demand. The trend of accumulating wealth is especially apparent in the Coastal West and the Gulf Coast. Over the past 20 years, wealth has grown at an annual average rate of 3.3% and 3.2% in the two regions, respectively. The economic impacts of a climate-related disaster on seaport systems are exacerbated by the concentration of wealth. As wealth accumulates, investments increase. Given the well-documented economic vitality of seaport systems, ports are a natural destination for new investments. When port operations are shut down due to a climate-related disaster, investors stand to lose billions in assets. Moreover, a 2008 OECD report found that five of the top ten ports globally in terms of assets exposed to coastal storm damage are located in the United States (Nicholls et al., 2008). These ports, along with the rest of the ports studied, held assets corresponding to 5% of global GDP at the time of publication. Asset exposure concentrated in ports could reach 9% of global GDP by 2070.

As port cities grow in both population and accumulated wealth, they face increasing risks of exposure to climate-related disasters. Climate scientists expect tropical cyclones and other storms to increase in frequency and intensity as the global climate warms (Lee et al., 2016; Bacmeister et al., 2018). With these expected increases, United States port systems face heightened risk profiles. The California tsunami study, for example, cited a storm with origin off

the coast of Alaska. Despite the distance between Alaska's off-shore fault lines and the Southern California coast, the high intensity of the expected storm is enough to yield heavy damages to the nation's economy. Beyond the direct effects of exposure to climate-related disasters, the higher risks associated with climate change allow insurance companies to charge higher premiums for coverage. Port authorities must maintain more robust disaster mitigation/resilience funds while simultaneously managing greater risk, higher value assets, and an increased workforce.

While port authorities take up this charge, society must work to adapt and respond to climate change in an effort to mitigate economic damages to and from risk-prone seaports. Science explains that the effects of climate change will only grow with time; Climate change will not subside without coordinated global action. In constructing such action, society must also confront a complex ethical quandary: how to work within the constraints of scarcity. Scarcity is the basic economic concern that there is a gap between limited resources and limitless wants. Society must allocate resources efficiently while recognizing that increased allocation of resources in one part of the economy demand reductions of resources in another part of the economy.

In terms of protecting United States seaports from the effects of climate change, ethical detractors may argue that it is immoral to redistribute resources from other pressing social issues. An increased effort to build climate resilience at the country's ports may take necessary resources away from those that need these resources elsewhere. As well, an effort to support the seaport industry and trade infrastructure may be seen as an attempt to bolster big business, especially at the expense of marginalized communities. These views may be held by those who do not believe in climate science or those who would prefer to follow partisan cues. They may also be held by those who care about climate change but see more visible need in the country's

housing markets or at the nation's borders. However, the economic damages associated with port degradation are so great that the transfer of resources must be justified. The destruction of the country's ports could yield staggering reductions in GDP, plunging the United States further into economic decline and creating a crisis that negates efforts to solve other issues such as homelessness and immigration.

As the threat of climate change grows stronger with time, society must decide how and with which resources it will respond. In the effort to mitigate trade-related damages, these debates will focus on whether ports should be protected by increasing resilience capabilities or whether climate change can be mitigated through a national reduction in carbon emissions and other environmental stressors. Regardless of the method employed, United States civil society must foster these debates and encourage action on climate change. Partisan floundering and general notions that climate change will not affect present-day generations are obstructive in nature and harmful to the nation's resilience capabilities.

As the United States seaport system accounts for \$4.6 trillion in economic activity per year, port protection is vital to the performance of the United States' economy. With prospects for more frequent and more intense tropical disasters due to climate change, each successive climate-related disaster raises the risk profile of United States ports. As risks increase, so do costs. Costs restrict productivity, and the economy at large is left to recover from decreased output and decreased welfare. Coordinated action can work to reduce or negate the effects of climate change on the national economy. Such action must begin at the ports. It is incumbent on the American public and its representatives in government to ensure that the United States' legacy on climate change is one of triumph.

References

- Bohringer, C., & Rutherford, T. (2017). U.S. withdrawal from the Paris Agreement: economic implications of carbon-tariff conflicts. Harvard Project on Climate Agreements. Retrieved from https://www.belfercenter.org/sites/default/files/files/publication/dp89_bohringer-rutherford.pdf
- Bacmeister, J. T., Reed, K. A., Hannay, C., Lawrence, P., Bates, S., Truesdale, J. E., ... Levy, M. (2018). Projected changes in tropical cyclone activity under future warming scenarios using a high-resolution climate model. *Climatic Change*, 146(3–4), 547–560. <https://doi.org/10.1007/s10584-016-1750-x>
- Batini, N., Eyraud, L., Forni, L., & Weber, A. (2014). Fiscal multipliers: size, determinants, and use in macroeconomic projections. *Technical Notes and Manuals*, 14(04), 1. Retrieved from <https://doi.org/10.5089/9781498382458.005>
- Becker, A. H., Acciaro, M., Asariotis, R., Cabrera, E., Cretegny, L., Crist, P., ... Velegrakis, A. F. (2013). A note on climate change adaptation for seaports: a challenge for global ports, a challenge for global society. *Climatic Change*, 120(4), 683–695. <https://doi.org/10.1007/s10584-013-0843-z>
- Carmichael, J. T., Brulle, R. J., & Huxster, J. K. (2017). The great divide: understanding the role of media and other drivers of the partisan divide in public concern over climate change in the USA, 2001–2014. *Climatic Change*, 141(4), 599–612. <https://doi.org/10.1007/s10584-017-1908-1>

Ehret, P. J., Van Boven, L., & Sherman, D. K. (2018). Partisan Barriers to Bipartisanship: Understanding Climate Policy Polarization. *Social Psychological and Personality Science*, 9(3), 308–318. <https://doi.org/10.1177/1948550618758709>

Joint Legislative Committee on Performance Evaluation and Expenditure Review (PEER). (2006). The impact of hurricane Katrina on Mississippi's commercial public ports and opportunities for expansion of the ports. Retrieved from <http://www.peer.state.ms.us/>

Kaplan, R. S. (2017). Impact of Hurricane Harvey and a Discussion of Key Structural Drivers Affecting U.S. Monetary Policy, 13.

Klotzbach, P. J., Bowen, S. G., Pielke, R., & Bell, M. (2018). Continental U.S. Hurricane Landfall Frequency and Associated Damage: Observations and Future Risks. *Bulletin of the American Meteorological Society*, 99(7), 1359–1376. <https://doi.org/10.1175/BAMS-D-17-0184.1>

Lee, C.-Y., Tippett, M. K., Sobel, A. H., & Camargo, S. J. (2016). Rapid intensification and the bimodal distribution of tropical cyclone intensity. *Nature Communications*, 7(1). <https://doi.org/10.1038/ncomms10625>

Manning, S. (2009). Is anything made in the U.S.A. anymore? You'd be surprised. *The New York Times*. Retrieved from <https://www.nytimes.com/2009/02/20/business/worldbusiness/20iht-wbmake.1.20332814.html>

Nicholls, R. J., Hanson, S., Herweijer, C., Patmore, N., Hallegatte, S., Corfee-Morlot, J., Chatua, J., & Muir-Wood, R. *Ranking Port Cities with High Exposure and Vulnerability to Climate Extremes: Exposure Estimates*. (2008). (OECD Environment Working Papers No. 1).

<https://doi.org/10.1787/011766488208>

Rose, A., Sue Wing, I., Wei, D., & Wein, A. (2016). Economic Impacts of a California Tsunami. *Natural Hazards Review*, 17(2), 04016002. [https://doi.org/10.1061/\(ASCE\)NH.1527-6996.0000212](https://doi.org/10.1061/(ASCE)NH.1527-6996.0000212)

Facts and Figures. (2017). *The Port of Los Angeles*. Webpage. Retrieved from <https://www.portoflosangeles.org/business/statistics/facts-and-figures>

Lee, C.-Y., Tippet, M. K., Sobel, A. H., & Camargo, S. J. (2016). Rapid intensification and the bimodal distribution of tropical cyclone intensity. *Nature Communications*, 7(1).

<https://doi.org/10.1038/ncomms10625>